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Numerical analysis of droplet breakup, cooling, and solidification during gas atomisation

Gezhou Wang[a], Yuanbin Deng [a], Frank Adjei-Kyeremeh [b], Jiali Zhang [a], Iris Raffeis [b], Andreas Bührig-Polaczek[b], Anke Kaletsch [a] and Christoph Broeckmann [a]

[a]Institute for Materials Applications in Mechanical Engineering (IWM), RWTH Aachen, Aachen, Germany; [b]Foundry Science and Foundry Institute (GI), RWTH Aachen, Aachen, Germany

ABSTRACT
As gas atomisation has been the main method for producing high-performance spherical powders in the past decades, its application in the production of metallic powders has become one of the main research subjects in this field. Since it is challenging to directly observe the atomising gas and to investigate melt flow states by experiments, numerical simulation is attracting increasing interest in studying the gas atomisation process. In this work, various computational fluid dynamics models were implemented to simulate the gas atomisation process. With the models, the droplet breakup, cooling, and solidification within the coupled process were investigated. The final mean particle size was predicted through numerical simulations and compared with the statistics extracted from the gas atomisation process, which shows that a reasonable mass median diameter of the particle can be predicted numerically. The results also show a clear relationship between the breakup trajectory and the resulting particle size.

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Gas atomisation; breakup; cooling; solidification; particle size and distribution; numerical methods; simulation

1. Introduction

Metal powder production industry has grown in the past decades due to the expanding fields of applications such as sintering, thermal spraying, isostatic pressing, hot forging, metal injection moulding and additive manufacturing [1–3]. In correspondence, the requirements for the geometric characteristics of metal powders like powder particle size, powder morphology and stability of different batches of products also drastically increased [4]. The gas atomisation process is considered as one of the main methods for the production of high-performance spherical metal powders with high productivity and suitability for a variety of materials [5]. In this process the high-speed gas jet and the high-temperature molten metal stream interact with each other under the conditions of extremely high cooling rate and momentum exchange [6]. The basic principle of gas atomisation is to transfer energy from the high-speed gas jet expanded through the nozzle to liquid metal, resulting in fragmentation and subsequently formation of metal droplets. The whole process can be divided into different stages. In the primary breakup, the melt undergoes film growth, ligament formation and parent droplets formation. Parent droplets undergo secondary breakup to form child droplets in different modes according to different Weber numbers. The child droplets then fly, spheroidize and solidify in the atomisation chamber, producing metal particles.

For the gas atomisation process, it is challenging and expensive to directly investigate and visualise the detailed atomising gas and melt flow states via conventional experimental methods. Therefore, numerical analysis with computational fluid dynamic (CFD) method has gradually become a major investigation tool. Several research works of single-phase flow numerical investigations were carried out to study the gas flow behaviour and the gas flow field, in which good agreement was observed between the simulation results and the experimentally observed phenomena [7–11]. However, these early studies could not successfully predict the gas atomisation process as they neglect the existence of the melt phase [7]. With the development of the CFD methods and the improvement of computing power, the melt phase was introduced in the simulation to enable the studies of breakup, cooling and solidification of the metal phase during gas atomisation. For the primary breakup, several numerical simulations [8–13] have been performed by using volume of fluid (VOF) multiphase method to study the melt morphology. Three primary breakup modes under different melt mass flow rates, i.e. the melt sheet mode, the hollow annular mode and the fountain mode, were summarised [14,15]. For the secondary breakup, several numerical simulations [16–19] were performed using discrete particle model (DPM) multiphase method to observe the droplet breakup behaviour and their spatial

CONTACT Yuanbin Deng y.deng@iwm.rwth-aachen.de Institute for Materials Applications in Mechanical Engineering (IWM), RWTH Aachen, 52062, Aachen, Germany

distribution in the gas flow field. For cooling and solidification, temperature changes of the droplets after secondary breakup were investigated [20–28]. In modern gas atomisation processes, secondary breakup as well as heat exchange between the droplets and the atomising gas happen simultaneously. However, in most of the current simulation approaches, the breakup process and the cooling and solidification process are considered as independent of each other. Compared to the experimental investigations of breakup behaviours by either taking high-speed images from the atomisation or x-ray attenuation, numerical simulation provides the opportunity to study these phenomena in a controlled way with much lower costs. This makes it especially important to understand the relationship between the process parameters and the final particle size. In this regard, effects of atomising gas pressure [16,18,19] and gas-to-melt mass flow rate ratio (GMR) [17–19] on the final particle size were studied, and the variable trends revealed by the numerical simulation results are in good agreement with the conclusions drawn from the experiments. However, how these two influencing factors affect the final particle size has not been further explored. Besides, the effect of the resulting changes in the gas flow field on the final particle size has also not been considered.

In this study, gas atomisation production of Inconel 718 powder was performed in a vacuum induction melting inert gas atomisation (VIGA) equipment. Based on this real production process, the melt breakup, cooling and solidification processes during gas atomisation were combined and studied by numerical simulations. The final particle size distribution was predicted through numerical simulations and compared with statistics obtained from experiments for further analyses. In addition, the influences of factors like the atomising gas pressure and the GMR on the final particle size were also studied. The gas flow field was introduced to study the influencing factors on the final particle size. Understanding of the whole gas atomisation process were deepened by analysing the evolution of the gas flow field.

2. Experimental and numerical methods

2.1. Materials and experimental methods

The gas atomisation process was performed in a gas atomisation equipment (VIGA2B, 2-litre Bottom Pouring, ALD Vacuum Technologies GmbH, Germany) which is operated in the Research Centre for Digital Photonic Production (RCDPP) at RWTH Aachen University. Nozzle structure and operating parameters of this equipment are shown in Figure 1 and Table 1.

A nickel-based super alloy Inconel 718 (IN718) (chemical composition in Table 2) was investigated and Argon was used as atomising gas. The gas atomisation process was carried out to collect the experimental process parameters which are listed in Table 3.

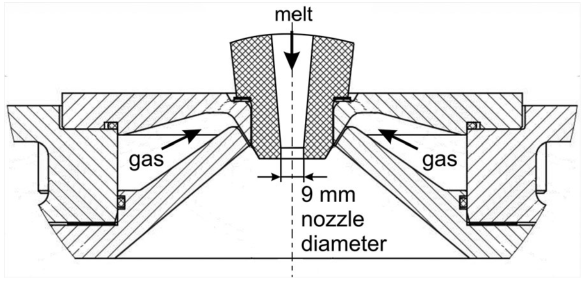

Figure 1. Nozzle structure of the gas atomisation equipment (Note that the nozzle is not shown in complete to reduce the size of the figure.)

After gas atomisation, the as-atomised powder was sieved to cut off flaky particles with a 315 µm sieve size. The powder was further classified using different sieves to characterise the powder between 90 and 315 µm, and the −90 µm size fraction was analysed with a dynamic imaging based particle size analyser (CAMSIZER X2, Retsch GmbH, Germany). In this way, the particle size distribution was determined for the total powder size range. The produced powder particles were hot embedded in a conductive resin that was ground and polished for the characterisations in a Helios NanoLab G3 CX DualBeam scanning electron microscope (SEM) (ThermoFisher Scientific, U.S.A.). For the embedded sample, grinding was conducted using an ATM Saphir360 with SiC papers from the mesh size of 180–1200. Subsequently, it was polished on a Struers Rotopol31 (Struers, Germany) with diamond suspensions of 6–0.25 µm on specific pads and cloths and thereafter with a colloidal silica (OPS) suspension of 50 nm particle size as the final step. Backscattered electron (BSE) and secondary electron (SE) images were taken at an accelerating voltage of 15 kV. Electron backscatter diffraction (EBSD) and energy-dispersive X-ray spectroscopy (EDS) measurements were performed using a Hikari XP II EBSD camera and an Octane EDS detector from Ametek-EDAX. The step size for EBSD measurements was 50 nm.

2.2. Mesh model with parameter setup

The physical model used in this study is based on the gas atomisation equipment which was introduced in

Table 1. Operating parameters of gas atomisation equipment.

Performance parameters (unit)	Value
Effective crucible capacity (litre)	2
Effective melt weight (kg)	approx. 14.4
Maximum melting temperature (K)	2173
Maximum melt mass flow rate (kg/s)	0.25
Atomising gas pressure (MPa)	1–4
Atomising gas temperature (K)	303–333
Maximum ultimate vacuum (Pa)	5
Maximum chamber pressure (Pa)	1.3×10^5

Table 2. Chemical composition of Inconel 718.

wt-%	C	Si	Mn	S	P	Cr	Ni(+Co)	Mo	Nb (+Ta)	Cu	Ti	Al	Fe
min.						17.0	50.0	2.8	4.75		0.65	0.20	Bal.
max.	0.08	0.35	0.35	0.015	0.015	21.0	55.0	3.3	5.50	0.3	1.05	0.80	

Table 3. Experimental process parameters of the gas atomisation process Inconel 718.

Experimental process parameters (unit)	Value
Atomising gas temperature after heating (K)	321.7
Atomising gas pressure (MPa)	2.99
Melt bath temperature (K)	1923
Melt inlet temperature (K)	1907
Melt mass flow rate (kg/s)	0.1
Melt chamber pressure (Pa)	1.063×10^5

section 2.1. The established mesh model is shown in Figure 2. Considering that the actual gas atomisation takes place in a small area at the bottom of the melt nozzle, the computational domain containing the atomising gas inlet with annular-slit structure, the melt delivery tube, and the atomising region at the bottom of the nozzle was selected as the mesh model for the simulation process. According to the sensitivity analysis, a simplified 3D model of a segment with rotational symmetry can ensure the accuracy of the results, preserve the gas flow characteristics in the circumferential direction, and greatly reduce the number of elements in the mesh to reduce the time required for the calculation. Therefore, the simplified periodic model with 15° angle was selected in this study. Based on the mesh independence study, 0.8 mm was selected as the maximum element size in the mesh model. The short mesh model with 80 mm length was used for the primary breakup process simulation, focusing on the region near the melt inlet. By shortening the length of the mesh model, the computational efficiency can be improved. The long mesh model with 800 mm length was used for the simulation of the coupling process. It is necessary to ensure that the melt particles undergo a complete solidification process in the computational domain. For this purpose, the mesh model needs to be of sufficient length.

Argon was used as the atomising gas, whose physical properties were obtained from ANSYS (2021R1) material database and listed in Table 4. The physical properties of the metal melt IN718 are shown in Table 5, the surface tension was determined according to the references [29,30], the dynamic viscosity was determined according to the reference [31], the other properties were determined according to the ANSYS material database and relevant references [32,33].

As for the boundary conditions, both the gas inlet nozzle and the melt inlet of the delivery tube were set as the pressure inlet boundaries. The outlet of the atomisation chamber, including the side and the bottom of the domain, were set as the pressure outlet boundary. The rest of the surfaces were set as the wall boundaries. The boundary conditions used for the numerical simulation are shown in Table 6.

2.3. CFD methods and models

Fluid flow is governed by three fundamental laws, which determine that mass, momentum, and energy must all be conserved during the process. Therefore, a set of time-dependent equations representing respectively the conservation of mass, momentum, and energy, were used to govern the fluid flow [34]. The differential forms of these equations are as follows.

$$\begin{gathered}\frac{\partial \rho}{\partial t}+\frac{\partial}{\partial x_i}(\rho u_i)=0\\ \frac{\partial}{\partial t}(\rho u_i)+\frac{\partial}{\partial x_i}(\rho u_j u_i)=-\frac{\partial p}{\partial x_i}+\frac{\partial \tau_{ij}}{\partial x_j}\\ \frac{\partial}{\partial t}(\rho E)+\frac{\partial}{\partial x_j}(\rho u_j H)=\frac{\partial}{\partial x_j}(u_i\tau_{ij})+\frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right)\end{gathered} \tag{1}$$

where ρ represents the fluid density, $u_i$and u_j represent the velocity components in the x_i and x_j directions, p the pressure, τ_{ij} the viscous stress tensor, E the energy,

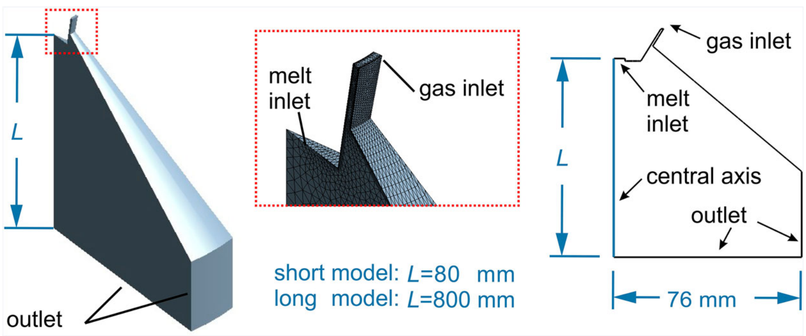

Figure 2. Mesh model for the simulation. The geometry of the model is magnified from the short model to the long model according to the ratio of the model length.

Table 4. Physical properties of Argon.

Properties (unit)	Value
Molecular weight ($kg{\cdot}mol^{-1}$)	28.0134
Dynamic viscosity ($kg{\cdot}m^{-1}{\cdot}s^{-1}$)	2.23×10^{-5}
Thermal conductivity ($W{\cdot}m^{-1}{\cdot}K^{-1}$)	0.017391
Specific heat ($J{\cdot}kg^{-1}{\cdot}K^{-1}$)	521.62
Density ($kg{\cdot}m^{-3}$)	Ideal gas

Table 6. The boundary conditions used for the simulation of numerical studies.

Wall	Boundary condition	Gauge pressure (Pa)	Temperature (K)
Gas inlet	Pressure inlet	2.99×10^6	321.7
Melt inlet	Pressure inlet	1.063×10^5	1907
Outlet	Pressure outlet	5	300

H the enthalpy $\rho E = p + \rho H$ and k the heat transfer coefficient of the gas flow.

Although the governing equations complete the description of the turbulent flow, it is still difficult or even impossible to obtain the analytical solutions or approximate analytical solutions in most cases. Therefore, discretization method was used to convert continuous models and equations into discrete counterparts to make it possible to obtain the numerical approximate solution of the governing equations. The finite volume method (FVM) is one of the discretization methods that are developing rapidly in the recent years. The basic idea of FVM is to divide the calculation area into grids and to make a non-repetitive control volume around each grid point. By applying the FVM, the differential forms of the governing equations are integrated over each control volume, resulting in a set of discrete equations. The FVM is characterised by a high computational efficiency and is widely used in the field of CFD at present.

To describe the turbulent flow of the fluid, i.e. to solve the governing equations, it is necessary to use the turbulence method in addition to the discretization method. Considering the complexity of the models involved, the computing resources at the current stage, and the computational accuracy of different turbulence models, the Reynolds stress model (RSM) was selected as the turbulence model. For RSM, the statistical analysis of turbulent flow requires that variables describing the flow are decomposed into a steady mean value and a fluctuating part.

$$\phi = \bar{\phi} + \phi' = \tilde{\phi} + \phi'' \tag{2}$$

where ϕ stands for each physical property of the fluid. The density and pressure are usually decomposed using the Reynolds approach, in which the steady mean value and the fluctuating part are represented by $\bar{\phi}$ and ϕ' respectively. Other flow variables such as velocity, temperature, enthalpy, and internal energy undergo a density weighted procedure, which is also known as the Favre method. In the Favre method, the steady mean value is represented by $\tilde{\phi}$ and the fluctuating part is represented by ϕ''. Therefore, the application of the averaging procedures to the differential forms of the governing equations yields the following set of equations.

$$\begin{aligned} &\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_i}(\bar{\rho}\tilde{u}_i) = 0 \\ &\frac{\partial}{\partial t}(\bar{\rho}\tilde{u}_i) + \frac{\partial}{\partial x_i}(\bar{\rho}\tilde{u}_j\tilde{u}_i) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j}(\tilde{\tau}_{ij} - \bar{\rho}\widetilde{u''_i u''_j}) \\ &\frac{\partial}{\partial t}(\bar{\rho}\tilde{E}) + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j\tilde{H}) = \frac{\partial}{\partial x_j}[\tilde{u}_i(\tilde{\tau}_{ij} - \bar{\rho}\widetilde{u''_i u''_j})] \\ &+ \frac{\partial}{\partial x_j}\left[k\frac{\partial \tilde{T}}{\partial x_j} - \bar{\rho}\widetilde{u''_j h''} + \widetilde{\tau_{ij}u''_i} - \bar{\rho}\widetilde{u''_j K}\right] \end{aligned} \tag{3}$$

where h represents the static enthalpy and K the turbulent kinetic energy [35].

The comparison with the governing equations reveals a new unknown quantity, which is called Reynolds stress tensor and can be directly computed in the RSM.

$$\tau_{ij}^R = -\bar{\rho}\widetilde{u''_i u''_j} \tag{4}$$

When considering the melt phase in the numerical simulation of the gas atomisation process, multiphase models were used to implement the interface between different phases. An interface resolving technique named volume of fluid (VOF) method was applied for the intact melt core and the dense spray regions that are present in the primary breakup process [36]. The discrete particle model (DPM) was used for simulating the dilute spray region that appears in the secondary breakup process.

Table 5. Physical properties of Inconel 718.

Properties (unit)	Value
Density ($kg{\cdot}m^{-3}$) [32, 33]	$8256.42 - 0.2019T - 1.3379 \times 10^{-4}T^2$
Dynamic viscosity ($kg{\cdot}m^{-1}{\cdot}s^{-1}$) [31]	$\log \mu = 2500/T - 3.68$
Thermal conductivity of liquid ($W{\cdot}m^{-1}{\cdot}K^{-1}$) [32, 33]	$4.95 + 0.0136T$ ($T > 1610$ K)
Thermal conductivity of solid ($W{\cdot}m^{-1}{\cdot}K^{-1}$) [32, 33]	$5.88 + 0.0165T$ ($T < 1528$ K)
Specific heat of liquid ($J{\cdot}kg^{-1}{\cdot}K^{-1}$) [32, 33]	775
Specific heat of solid ($J{\cdot}kg^{-1}{\cdot}K^{-1}$) [32, 33]	600
Surface tension ($N{\cdot}m^{-1}$) [29, 30]	1.88
Solidus temperature (K) [32, 33]	1528
Liquidus temperature (K) [32, 33]	1610
Latent heat ($J{\cdot}kg^{-1}$) [32, 33]	227000

In the VOF method, an indicator γ was introduced to identify the gas-to-melt interface, and a set of momentum and energy conservation equations for both melt and atomising gas phases was solved.

$$\gamma_m = \frac{\Omega_m}{\Omega}, \quad \gamma_g = \frac{\Omega_g}{\Omega} \tag{5}$$

with

$$\gamma_m + \gamma_g = 1 \tag{6}$$

where the control volume Ω is also named as the computational cell and determined by the division of the mesh, the subscript m denotes the melt phase, the subscript g denotes the gas phase. The melt volume fraction γ_m is a numerical property for the identification of melt and gas phases, which is indicated by using the values between 0 and 1 in a control volume, as shown in Equation (7).

$$\gamma_m = \begin{cases} 1 & \textit{only melt phase} \\ 0 < \gamma_m < 1 & \textit{two phases} \\ 0 & \textit{only gas phase} \end{cases} \tag{7}$$

The physical properties of the fluid can be computed as follows.

$$\phi = \gamma_m \phi_m + \gamma_g \phi_g \tag{8}$$

The investigation was carried out by the application of the VOF model, which solves a single set of momentum and energy equations for both phases. This was achieved by monitoring the volume fraction of the melt and gas throughout the computational domain, using a continuity equation (9) for volume fraction. Depending on the volume fraction γ weighted value of each phase, the density ρ and viscosity u are calculated by equations (10)–(11).

$$\frac{1}{\bar{\rho}_q}\left[\frac{\partial}{\partial t}(\tilde{\gamma}_q \bar{\rho}_q) + \frac{\partial}{\partial x_i}(\tilde{\gamma}_q \bar{\rho}_q)\right] = 0 \tag{9}$$

$$\bar{\rho} = \tilde{\gamma}_m \bar{\rho}_m + (1 - \tilde{\gamma}_m)\bar{\rho}_g \tag{10}$$

$$\bar{u} = \tilde{\gamma}_m \bar{u}_m + (1 - \tilde{\gamma}_m)\bar{u}_g \tag{11}$$

In DPM, the gas phase is regarded as a continuous phase, while the melt droplets are regarded as discrete particles. Motion of the particles is governed by Newton's second law based on the force balance.

$$m_p \frac{d\boldsymbol{u_p}}{dt} = \frac{1}{2}\rho_g C_D |\boldsymbol{u_g} - \boldsymbol{u_p}|(\boldsymbol{u_g} - \boldsymbol{u_p})A_p + (\rho_p - \rho_g)\boldsymbol{g}V_p \tag{12}$$

where $\boldsymbol{u}$ represents the velocity vector, the subscript p denotes the particle, C_D represents the drag coefficient, $\boldsymbol{g}$ the gravitational acceleration, A_p the particle projected area and V_p the particle volume. Therefore, the particle velocity $\boldsymbol{u_p}$ can be calculated and the new particle position can be updated as follows.

$$\boldsymbol{x_p^{i+1}} = \boldsymbol{x_p^i} + \boldsymbol{u_p^{i+1}}\Delta t \tag{13}$$

The continuous gas phase affects the movement of the discrete phase particles through force. Meanwhile, the particles also have influences on the continuous gas phase. Therefore, there is the coupling between different phases. This two-way coupling approach requires that the momentum source of a particle is loaded into the momentum equation of the fluid in the control volume of the particle within the current time step. The momentum exchange source term $\boldsymbol{S_p}$ follows the following equation.

$$\boldsymbol{S_p^{i+1}} = -\frac{1}{\Omega \Delta t}\sum m_p(\boldsymbol{u_p^{i+1}} - \boldsymbol{u_p^i}) \tag{14}$$

The effect of the gas flow field on the particles not only causes the movement of the particles, but also leads to further fragmentation of the particles, which is the main mechanism in the secondary breakup process. The hybrid Kelvin-Helmholtz Rayleigh-Taylor (KHRT) breakup model that was used to describe the secondary breakup process is formulated based on the theoretical analysis of the stability of the liquid–gas interfaces, when they are accelerated in the normal direction of the liquid plane [37]. It was assumed that there is a liquid core in the area close to the nozzle, from which the child droplets break off and experience a sudden acceleration as they are ejected into the freestream. In the KHRT breakup model, the 'Kelvin-Helmholtz instability' (KH) was combined with the 'Rayleigh-Taylor instability' (RT), and a droplet breaks up by the model that predicts a shorter breakup time [38]. For the gas atomisation process, the KH instability is considered as the main effect within the liquid core, while RT instability becomes the dominant effect for further downstream region. Therefore, the final child droplet size was determined by the RT instability, as shown in the following:

$$r_{child,RT} = \Lambda_{RT}\frac{C_{RT}}{2}\left[\frac{\rho_p - \rho_g}{\rho_p + \rho_g}\right]^{1/3} \tag{15}$$

with

$$\Lambda_{RT} = 2\pi\sqrt{\frac{3\sigma_p}{|\boldsymbol{F}|(\rho_p - \rho_g)}} \tag{16}$$

$$|\boldsymbol{F}| = \frac{3}{8}C_D\frac{\rho_g u_{rel}^2}{\rho_p r_p} \tag{17}$$

where Λ_{RT} represents the fastest growing wavelength defined by RT breakup models, C_{RT} represents the breakup radius with a constant value of 0.1, σ_p represents the particle surface tension coefficient, $\boldsymbol{F}$ represents the drag force acting on the particles, u_{rel}

represents the relative velocity between gas and particles, r_p represents the particle radius before breakup, which is also referred to as parent droplet radius.

When the particles move and break up in the gas flow field, they transfer heat to the surrounding environment. Convection is the main form of heat exchange between the particle and the surrounding atomising gas environment. The influence of thermal radiation on particle temperature can be ignored due to the rapid solidification condition [38]. Considering the high thermal conductivity of the metal melt, each particle can be regarded as temperature constant. Therefore, Newton's law of cooling was selected as the cooling model. The temperature of the particles can be calculated using the following equation:

$$m_p \frac{dH_p}{dt} = S_p h(T_g - T_p) \tag{18}$$

where H_p represents the particle enthalpy, S_p the surface area of the particle, h the convective heat transfer coefficient, T_g the temperature of the atomising gas and T_p the temperature of a particle.

In summary, numerical studies were performed to model a gas atomisation process which consists of two stages. The first stage corresponding to the primary breakup process was studied using RSM turbulence model and VOF multiphase method. In this stage, numerical analyses of the gas flow field and the melt morphology were carried out. Information on the flow state of the droplets after the primary breakup was summarised and set as the initial condition for the simulations in the next stage. In the second stage, the coupling process of the droplet breakup and the solidification were studied using RSM turbulence model, DPM, KHRT breakup model and the cooling model. Within this coupling process, numerical analyses of the gas flow field, the melt morphology, the final particle size and distribution, and the influencing factors affecting the final particle size were carried out.

In total two mesh models with the same nozzle structure but different lengths were established and used in the numerical studies of the two stages, as shown in Figure 2. The 'length of the model' was defined as the distance from the melt inlet to the bottom outlet. The short mesh model with 80 mm length was used for the simulation of the primary breakup process, focusing on the region near the melt inlet to improve the computational efficiency. The long mesh model with 800 mm length was used for the simulation of the coupling process to ensure that the melt droplets undergo complete cooling and solidification processes through the computational domain.

3. Results and discussion

3.1. Gas flow field analysis and melt morphology during the primary breakup process

Figure 3 shows the velocity contour of the gas flow field and the streamline with vector arrows in the recirculation zone. The atomising gas is ejected from the gas inlet at a pressure of 2.99 MPa, forming a supersonic under-expanded gas jet along the melt tip boundary. The under-expanded gas jet is subjected to constant cycles of compression-expansion-compression along the downstream direction, forming shock wave of the cell structure. The gas jet divides the flow field within the atomisation chamber into two areas, namely the high-speed area and the free boundary of the edge, as separated by the white dashed line in Figure 3. The high-speed area locates near the central axis, with the free boundary of the edge surrounding at the outside. The gas flow along the melt tip separates as it attempts to turn at the melt tip

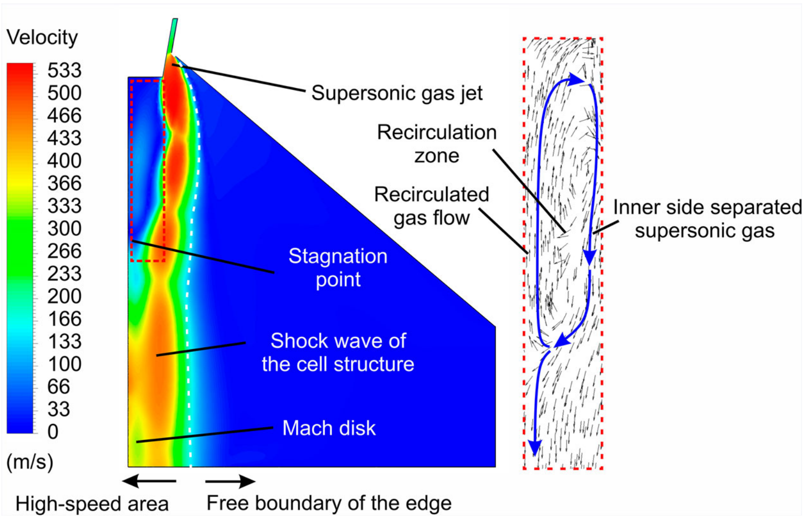

Figure 3. The velocity contour of the gas flow field and streamlines with vector arrows in the recirculation zone.

base corner. The inner side separated supersonic gas flow continues to expand as it moves towards the central axis, forming the recirculation zone at the bottom of the melt delivery tube. The inner gas flow near the central axis of the recirculation zone has an upward intermediate gas flow velocity such that its kinetic energy is sufficient to counteract the resistances exerted by the incoming fluid from the melt nozzle and the internally recirculated gas flow along the central axis towards the melt tip. After a certain distance, the internally recirculated gas flow deflects to the surrounding under the action of the fluid entering from the melt nozzle, generating a radial velocity, while the axial velocity decreases to zero. Therefore, a radial gas flow forms along the tip of the melt nozzle and interacts with the inlet atomising gas jet at the edge of the melt tip. Under the action of the supersonic gas jet flow, a downward axial flow velocity is generated for the gas flow outside of the central recirculated gas flow and surrounded by the supersonic gas jet which flows in the direction away from the nozzle. The large velocity gradient between the inner upward central recirculated gas flow and outer supersonic gas jet flow generates a strong mixing layer. A part of the gas flow near the outer interface of the mixing layer escapes from this area under the shear action with the supersonic gas jet. Another part of the downflow gas near the inner interface of the mixing layer gathers towards the central axis and re-enters the recirculation zone. These two parts of the gas are separated at the end of the recirculation zone, which is named as the stagnation point as labelled in Figure 3. In the downstream area of the stagnation point, the gas flow rapidly accelerates to be supersonic, then decelerated to be subsonic, and again gradually accelerates in fluctuations during the subsequent flow. A shock wave normal to the central axis can be observed, which is called the Mach disk and locates at the velocity minimum in the local area. A comparison with similar research studies [39–42] on the gas flow field during the gas atomisation process shows a good agreement on the numerically simulated gas flow behaviour and the structures in the gas flow field.

The melt flows from the inlet (Figure 4) under the action of gravity, the aspiration pressure and the radial pressure gradient formed at the bottom end of the melt nozzle from the gas flow field. The existence of the pressure gradient forces the melt out of the nozzle to flow radially outward, forming a liquid film at the tip of the melt nozzle. The melt is in direct contact with the supersonic gas jet along the melt tip boundary. After intense energy exchange, the edge part of the melt film is continuously peeled off and broken into strips or large droplets which then go downstream. According to the primary breakup modes summarised by Mates and Settles [14], it was found that the simulation results of the primary breakup

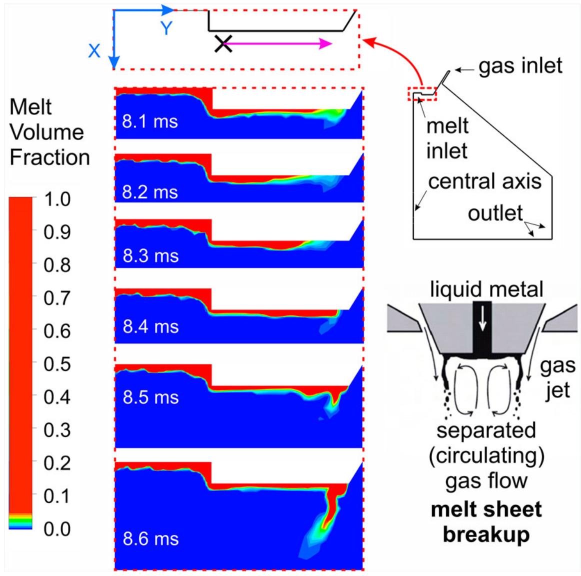

Figure 4. Melt morphology within the primary breakup process and injection of the parent droplets in the coupling process of the droplet breakup and the solidification.

state of the melt in this study correspond to the melt sheet mode.

Many pre-existing works [16–18,43,44] assume that parent droplets are released at the top corner of the melt nozzle without an initial velocity. However, this setting ignores the connection between the primary and the secondary breakup processes. Such an oversimplification can lead to inaccurate simulation results. Therefore, in this study, based on the analysis of the melt flow during the primary breakup process, the initial conditions of the injected parent droplets, including positions, velocities, and sizes, in the coupling process were determined. Figure 4 shows the position and moving direction of the injected parent droplets in the initial state of the coupling process. Note that the scale bar of the melt volume fraction in the figure was adjusted in a special way for the best visualisation of the melt morphology. Considering that the area where the primary breakup occurs, the position of the black cross was selected as the injection position of the parent droplets in the coupling process, which is 6 mm away from the central axis in the y-direction and 2 mm away from the melt inlet in the x-direction. The velocity was set as 6 m/s, and the direction of the velocity is indicated by the pink arrow, which is perpendicular to the central axis and points outward. As the initial droplets of the secondary breakup, the size of the injected parent droplets at the exit after primary breakup was assumed as between 10-100% of the nozzle diameter [45]. Therefore, the diameter of the injected parent droplets was set as 50% of the melt nozzle diameter according to its maximum occurrence probability. All information about the initial condition of the coupling process is listed in Table 7.

Table 7. The initial condition of the coupling process.

Initial condition		Setup
Injected parent droplets	Position	x = 2 mm y = 6 mm
	Velocity	6 m/s in y-direction
	Diameter	4.5 mm

3.2. Gas flow field analysis and melt morphology within the coupling process

Figure 5 shows the velocity contour of the stable gas flow field and the spatial distribution of the droplets with different sizes within the coupling process. In fact, the choice of different time points has little influence on the simulation results. In Figure 5, the distribution of the droplets with different diameters can be distinguished by different colours. Considering the great difference in droplet sizes before and after the secondary breakup, the colour scale was selected as logarithmic coordinate system. The size and number of the droplets shown in the figure are adjusted by scaling. It can be directly seen that the injected parent droplets are constrained in a conical shape space near the upper surface of the atomisation zone. The droplet sizes in the central area of the cone are larger than that close to the outer shell, while the initially injected parent droplet sizes stay unchanged. In the high-speed gas flow area, the droplet trajectories are significantly deflected, and the droplet sizes change significantly during the trajectory deflection process, reducing to 200–400 μm. It can be observed from the droplet behaviour that there are mainly two kinds of motion trajectories of the injected parent droplets. On the one hand, some of the parent droplets are broken up within a short period of time after injection, moving outward along the conical surface, and deflecting in the high-speed gas flow area without further reduction of their sizes. On the other hand, the rest of the parent droplets in the central area of the cone are protected by the outer droplets to keep their sizes and temperatures basically unchanged. The deflection occurs in the high-speed gas flow area, and the droplet sizes decrease with the change of their moving directions. The droplets move along both kinds of trajectories from the recirculation zone through the shear layer to the high-speed gas jet zone, where they converge and move further downstream together. The sizes of the broken droplets near the nozzle are much larger than the final particle sizes, indicating that the droplets are broken again during the subsequent movement. This means that the droplets show multiple breaking in the flow field and gradual decreasing of their diameters.

Figure 5 also shows that some main characteristic structures of the flow field still exist, including the recirculation zone, the Mach disk, and other structures. Compared with the gas flow field without considering the melt phase (Figure 3), it was found that the flow field changed significantly. The maximum velocity in the gas flow field and the velocity of the gas flow near the central axis decrease due to the interaction with the droplets. The outer high-speed gas flow fluctuates due to the random distribution of the droplets. Therefore, it is believed that the strong interaction between the gas and the droplets makes it necessary to consider the two-way coupling approach when investigating the secondary breakup in the coupling process.

3.3. Droplet breakup, cooling and solidification

Figure 6 shows the change of the temperature and diameter of droplets in the coupling process. Figure 7 shows the temperature difference of the droplets between the beginning and the end of the solidification. Droplets in the T-t charts in Figure 6(a) are represented by '+' and coloured according to different

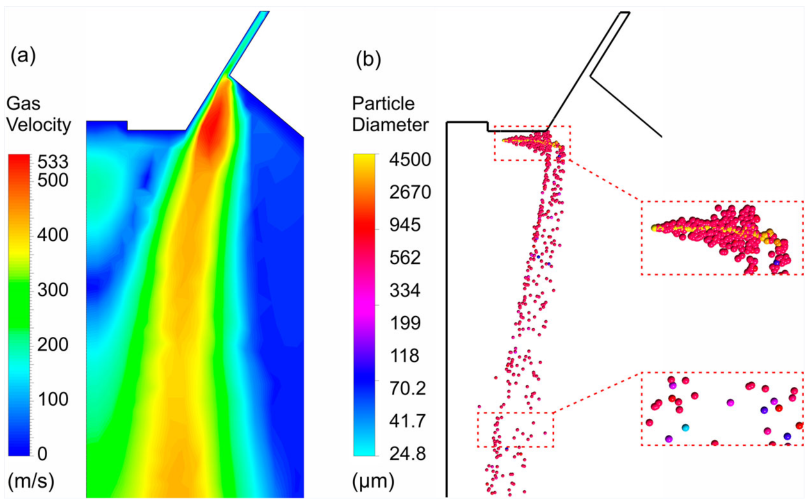

Figure 5. The velocity contour of the gas flow field (a) and spatial distribution of the droplets with different sizes (b) within the coupling process (t = 20 ms).

diameters for better visualisation of the temperature changes of the droplets with the same size and with different sizes. Droplets in the Figure 6(b) are represented by '×' on a logarithmic scale to better show the distribution of different droplet diameters. The size of the droplet markers shown in both graphs has been scaled for better presentation.

Figure 6(a) reveals many droplets with various colour markings representing different diameters in 5 ms of secondary breakup. The temperatures of those droplets are high in the initial stage and then decrease rapidly from 1907 K to 1610 K. As the temperature decreases to a characteristic temperature range between 1610 and 1528 K, the droplet temperature change slows down. However, the particle temperature change jumps to the initial fast rate again, thereafter, continues to slow down as the temperature further decreases with increasing time. The comparison of the colours of the particles shows that the temperature changes are similar for the particles with similar diameters, whereas the temperature changes are significantly different for particles in different size ranges. In addition, some of the parent droplets marked in red have almost a constant temperature in the time range from 0 to 1.2 ms. Furthermore, the temperature of many droplets which are distributed within the diameter range of 25–32 μm are kept constant after 0 ms.

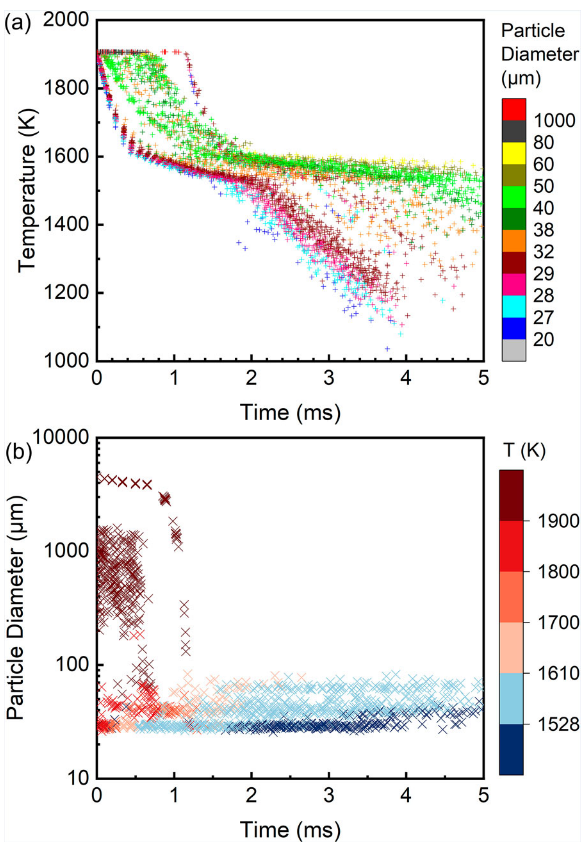

Figure 6. Change of temperature (a) and diameter (b) of droplets in the coupling process.

Figure 6(b) shows that the droplet size between 0 and 2 ms follows a large size reduction. Droplets after solidification and size reduction have diameters in the range of 25–80 μm. The breakup time of the initial injected parent droplets is up to 1.2 ms. After 1.2 ms, the particle size distribution no longer changes appreciably. Seen from the colours of points in the figure, the temperature of the droplets decreases continuously during the breakup process. Particles with temperature below 1528 K appear for the first time after 1.5 ms.

Summarising the information given above, the initial parent droplets with diameters around 4.5 mm are injected into the gas flow field at 0 ms. They subsequently break up to generate child droplets which are in the diameter range of 25–80 μm. During solidification as shown in Figure 7, the cooling rate of the droplets significantly reduces due to the generation of latent heat. When the droplet temperature is below the solidus temperature, the melt completely solidifies to form particles. As the particle temperature decreases, the heat exchange rate between the particles and the gas phase decreases, resulting in a significant reduction in the cooling rate. The temperature changes of particles with different diameters over the time are significantly different. The particles with large diameters show a relatively slower temperature change and higher temperature than the small particles, which might be resulted from the fact that small particles have faster heat transfer in the gas flow field. However, for all particles with different diameters, their solidification time is greater than the breakup time, and the melt has sufficient time to complete the full breakup before solidification. In this case, the breakup of the droplets is not affected by the solidification, and it can be considered that the droplet break-up is completed before freezing.

3.4. Prediction of particle size distribution

The particle size distribution (PSD) as simulated and experimentally obtained are highlighted and

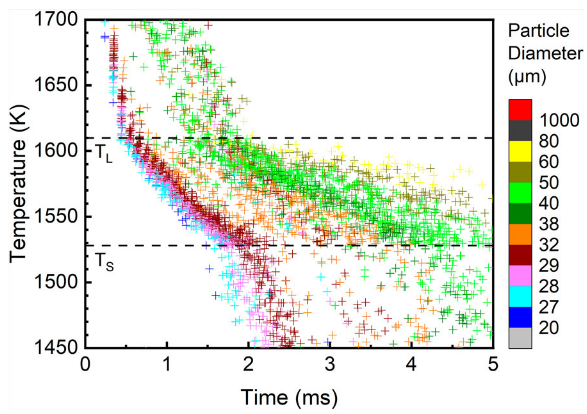

Figure 7. The temperature change of droplets between the liquidus and solidus temperatures, T_L and T_S.

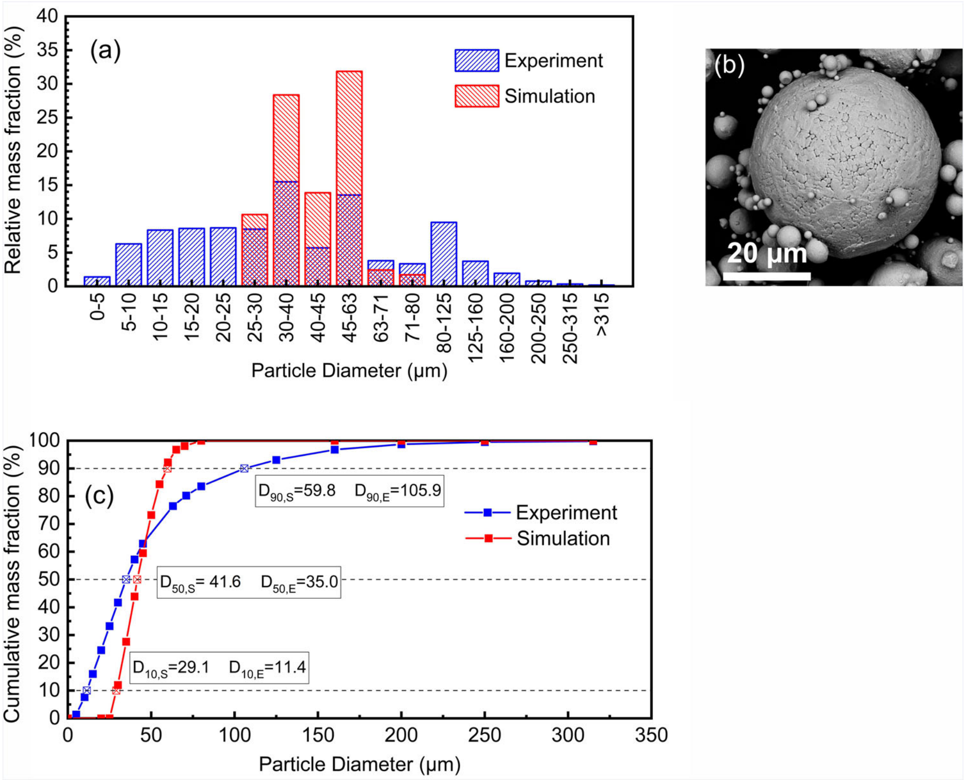

Figure 8. Particle size distributions of experimental and simulation results (a) and (c); SEM image of powder particles (b).

compared in Figure 8(a). Few satellite particles as well as some non-spherical particles can be observed in the SEM as shown in Figure 8(b). The particles are mostly spherical, and the average particle sphericity, symmetry, and aspect ratio are 0.81, 0.93 and 0.85, respectively. The surface morphology of the particles observed in the SEM shows inter-dendritic structures, as can be seen in Figure 8(b).

By analysing the simulation results of particle size distribution, relative and cumulative mass fractions were obtained and compared with the experimental results, as shown in Figure 8. Furthermore, three particle size eigenvalues from numerical simulation were also obtained and compared with those from experiments, as listed in Table 8. From these figures and tables, it can be observed that the mass median diameter of the particles (D_{50}) from simulation agrees well with that obtained experimentally; the D_{10} from the simulation is larger than that from the experiments; the simulated D_{90} is smaller than that of the experimental result. This indicates that the simulated particles have a relatively consistent D_{50} with the experimental results, but the particle distribution is significantly narrower. This confirms that a reasonable D_{50} of the particles can be obtained based on the applied numerical simulation method, whereas the simulated particle size distribution may not be reliable.

Table 8. Particle size eigenvalue of experimental and simulation results.

Particle size eigenvalue	Experimental results (μm)	Simulation results (μm)
D10	11.4	29.1
D50	35.0	41.6
D90	105.9	59.8

The following inferences can be made for the similarities and differences between the simulation and the experimental results. As uniform injected parent droplets were set in the simulation, the randomness in the particle size, position, velocity and direction caused by the primary breakup were automatically neglected. In this way, the randomness of droplet breakup was weakened to a certain extent, which could result in a narrow particle size distribution. Similar inferences were also given in the literature [19]. For the mass median diameter of the particles, 50% of the nozzle diameter was selected as the pre-set value of the injected parent droplet diameter, and the position and velocity of the injected parent droplets were also selected as their maximum probability values. Following this, the simulated particle size was assured to be in a reasonable distribution range. However, the relevant information of the particles was simplified during the simulation process which can result in the reduction of the breakup randomness.

To improve the accuracy of the particle distribution in the simulation, the droplet size and velocity after the primary breakup were introduced as variables in the coupling process. 10% and 100% nozzle diameters (0.9 and 9.0 mm, respectively) were also selected as the minimum and maximum diameters of the injected parent droplets. Assuming that the droplet velocity is zero when detaching from the melt column, only the drag force exerted by the gas flow is applied on the parent droplets, while other forces are ignored. Equations of the drag force and the conservation of energy are listed below.

$$F = \frac{1}{8} C_D \pi d_p^2 \rho_p u_{rel}^2 \tag{19}$$

$$F \cdot s = \frac{1}{12} \pi d_p^3 \rho_p u_p^2 \tag{20}$$

where d_p is the parent droplet diameter, F the drag force acting on the particles, u_{rel} the relative velocity between gas and particles, C_D the drag coefficient, u_p the particle velocity, ρ_p the density of the particle, s the distance to the injection point. From the equations, the drag force is proportional both to the square of the parent droplet diameter and to the square of the relative motion velocity. Since the droplet velocity is much smaller than the gas flow velocity in this process, it can be assumed that the relative velocity between gas and particles equals to the gas flow velocity which is a constant determined by the inject gas pressure. According to the equations (19) and (20), droplets with different sizes move at the same distance to the injection point with different velocities under the drag force of the gas. Therefore, the relationship between the diameter and the velocity of the injected parent droplets can be obtained, as shown in the equation.

$$d_p u_p^2 = const. \tag{21}$$

Given that the inject velocity of droplets with 4.5 mm diameter is 6 m/s, it can be calculated that the inject velocity of droplets with 0.9 mm diameter is 13.4 m/s, and that with 9.0 mm diameter is 4.2 m/s. Both the diameters and the calculated different velocities of the injected parent droplets were set as initial conditions in the coupling process.

The simulation results were compared with the previous experimental results, as listed in Table 9 and plotted in Figure 9. It can be found that when both, variations of the diameter and the velocity of the injected parent droplets, are considered, the simulated particle size exhibits a wider distribution, which is closer to the experimental results of the particle size distribution. However, since there are still many variables not considered in the simulation process, a certain gap between the simulated particle size distribution and the experimental one still exists. Possible influencing parameters include the velocity direction and the position of the injected parent droplets as well as the existence of small sized droplets with large velocities or large sized droplets with small velocities. In addition, due to the lack of the distribution of different parent particle sizes after primary breakup, it is impossible to make relevant settings for the particle size distribution of the injected parent droplets in the coupling process. Therefore, the discussion of the rationality and accuracy of the simulated particle size distribution can only be limited to its coverage without extending to other particle diameter range.

It was surprising to observe that the final particle sizes decrease when the sizes of the injected parent droplets increase, as it was generally reported in the literature [43,46] that injected parent droplet size has no significant influence on the final particle sizes. As aforementioned, the initial velocity of the parent droplets was normally taken as 0 or very small in the reported simulation works, however, moderate or high velocities of the droplets were implemented in the simulation in this work. This might be the reason for the drastically different observations between the literature data and

Table 9. Particle size distribution for experimental and simulation results with different injected parent droplets diameters and velocities.

Particle diameter range (μm)	Experimental cumulative mass fraction (%)	Simulated cumulative mass fraction (%) (1) 0.9 mm 13.4 m/s	(2) 4.5 mm 6 m/s	(3) 9.0 mm 4.2 m/s
0–5	1.39	–	–	–
5–10	6.28	–	–	–
10–15	8.33	–	–	32.92
15–20	8.58	–	–	25.04
20–25	8.67	–	0.04	18.41
25–30	8.48	–	11.97	11.51
30–40	15.48	–	31.90	9.23
40–45	5.70	–	15.59	1.50
45–63	13.53	–	35.85	1.11
63–71	3.80	–	2.72	0.28
71–80	3.34	0.10	1.93	–
80–125	9.48	88.05	0.04	–
125–160	3.72	11.85	–	–
>160	3.22	–	–	–

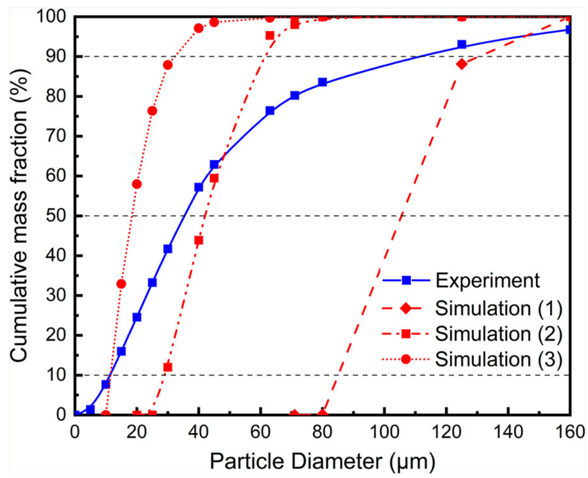

Figure 9. Particle size distribution of experimental and simulation (1-3) results. The inject parent droplets with a diameter of 0.9 mm and a velocity of 13.4 m/s were defined in the simulation (1), with a diameter of 4.5 mm and a velocity of 6 m/s in the simulation (2), and with a diameter of 9.0 mm and a velocity of 4.2 m/s in the simulation (3).

the data shown here. The data of Zeoli and Gu [43] and Zou and Xiao [46] were extracted and listed together with the results obtained in this study in Table 10 for a close comparison. Seen from this comparison, it was speculated that the relationship between the final particle size and the injected parent droplet size is affected by the breakup trajectory under the influence of the melt mass velocity. Small sized injected parent droplets are light and therefore brought further by the gas in the radial direction, so that the breakup trajectory of small sized droplets is located outside that of the large sized droplets.

Under different velocity conditions, the trajectories of parent droplets with different sizes may locate differently in the gas flow field, which can strongly influence the final particle sizes. By separating the droplet sizes to small and large with 4.5 mm and dividing the velocity into two groups, small or 0 and moderate or high, 4 distinct trajectories are shown in Figure 10 and described in the following. In the general cases reported in the literature, the velocity is small or 0, therefore, the breakup trajectories of all droplets are located in the recirculation zone or close to the high-speed gas jet area (dashed black and grey arrows). In this situation, although there is a larger relative velocity difference between the small sized droplets and the gas than that between the large sized droplets and the gas, the small droplets cannot be broken up to a high degree due to their short breakup time (shown in Figure 6). Therefore, the final particle size is not significantly influenced by the injected parent droplet size. However, in our study the velocity of the parent droplets is moderate or high, which makes it possible that these droplets are brought further into the high-speed gas jet area and follow the breakup trajectories shown by the solid black and grey arrows shown in Figure 10. In this case, due to a very short breakup time, the breakup of the small droplets stays like the droplets with 0 or small velocity. However, the extremely large relative velocity difference between the large droplets and the gas at their locations strongly intensifies their breakup. Therefore, the final particle size shows an opposite trend to the injected parent droplet size, which is that the larger the injected parent droplet size, the smaller the final particles.

Table 10. Relationship between the final particle size and the injected parent droplet size.

Case	d_p (mm)	D50 (μm)	Relationship	Breakup trajectory in Figure 10
This study	0.9	95.6	Injected parent droplet size increases, D_{50} decreases	Solid black and grey arrows
	4.5	41.6		
	9.0	18.4		
Zeoli and Gu [43]	1.0	32.0	No significant change in D_{50}	Dashed black and grey arrows
	3.0	39.9		
	5.0	41.0		
Zou and Xiao [46]	1.5	71.4		
	2.0	74.0		
	2.5	72.0		
	3.0	73.6		

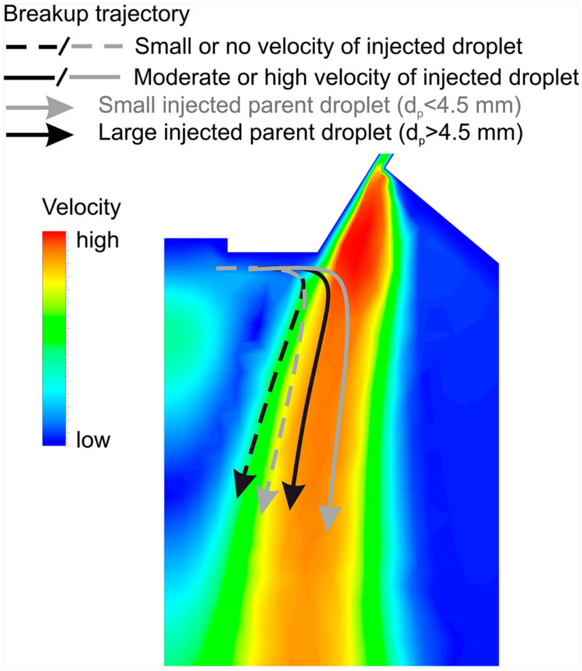

Figure 10. Possible breakup trajectories of injected parent droplet with different sizes and velocities.

3.5. Influencing factors on the final particle size and powder microstructure

The quality of powder as obtained after gas atomisation is dependent on atomisation parameters such as

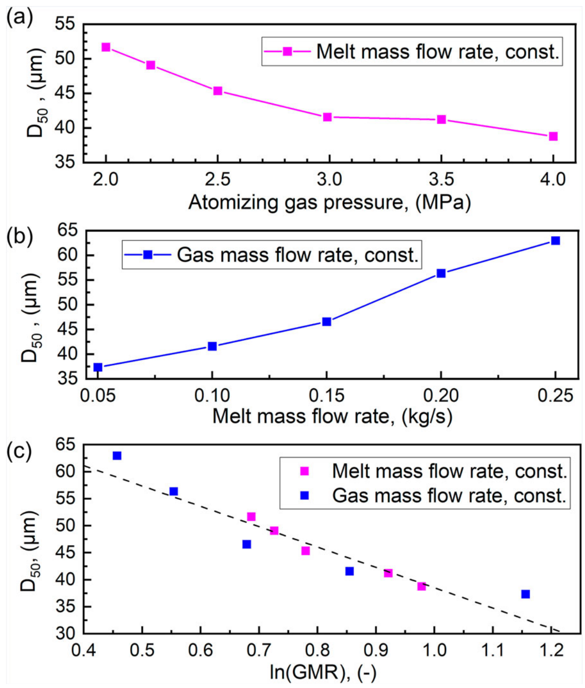

Figure 11. Calculated D_{50} for different atomising gas pressures (a), for different melt mass flow rates with a constant gas mass flow rate (b), and for different GMRs (c).

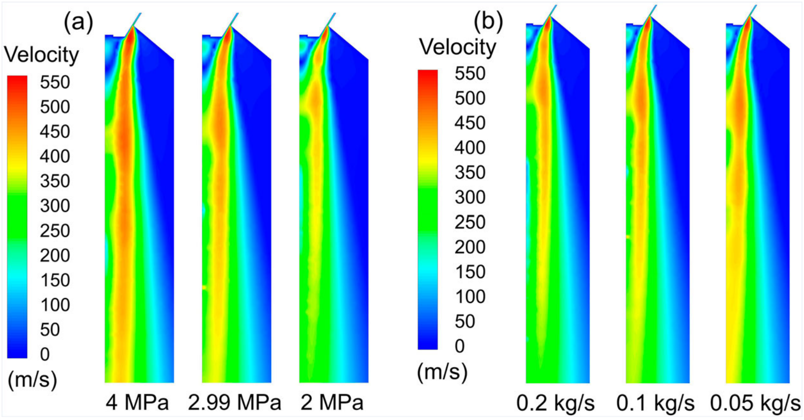

Figure 12. Gas flow fields for different atomising gas pressures (a) and melt mass flow rates with a constant gas mass flow rate (b)

gas pressure, gas flow rate and gas to melt ratio. The gas pressure is known to have the most significant influence on the particle size distribution, as too low gas pressures may lead to unsolidified droplets ending up in the powder collecting can, whereas extremely high pressures may lead to freezing of melt at the nozzle [47]. When the gas pressure increases, the gas mass flow and velocity also increases, leading to a longer gas plume which favours a more effective secondary breakup and the formation of finer particles [48,49]. The experimentally performed atomisation used a gas pressure of 2.99 MPa which resulted in a D_{50} of 35 µm. In the simulation, the gas atomisation pressure varied between 2 and 4 MPa to ascertain its influence on the mean particle diameter as shown in Figure 11 (a). Consistent with the findings from other works [49–51], the D_{50} was found to decrease with increasing gas pressure (Figure 11(a)). Simulation with a gas pressure of 2.99 MPa (the same as in the experiment) resulted in a D_{50} of 41.57 µm that is also close to the experimentally obtained D_{50} value of 35 µm.

When the gas-to-melt mass flow rate ratio (GMR) increases, atomised droplets are also accelerated, which leads to a decrease in final particle size [50,51]. Under a constant gas pressure, the mean particle diameter increases linearly with the melt mass flow rate because of the reduced impact of the gas pressure and its attendant slower acceleration of the droplets. This phenomenon was also observed in the simulation where the melt mass flow rate was varied between 0.05 and 0.25 kg/s with a constant gas mass flow rate and the D_{50} was found to increase from 37.33 µm to 62.96 µm (Figure 11(b)).

Based on the data from Figure 11(a) and (b), the GMRs were calculated. Calculated D_{50} with respect to different GMRs were plotted in Figure 11(c). The relationship between the GMR and D_{50} which can be given as $D_{50} \propto \frac{1}{\sqrt{GMR}}$ fits well, when the GMR varies in the range of 3-10. When the GMR is too high, the final particle sizes cannot continue to decrease with increasing the GMR. One reason for this can be that the primary breakup mode

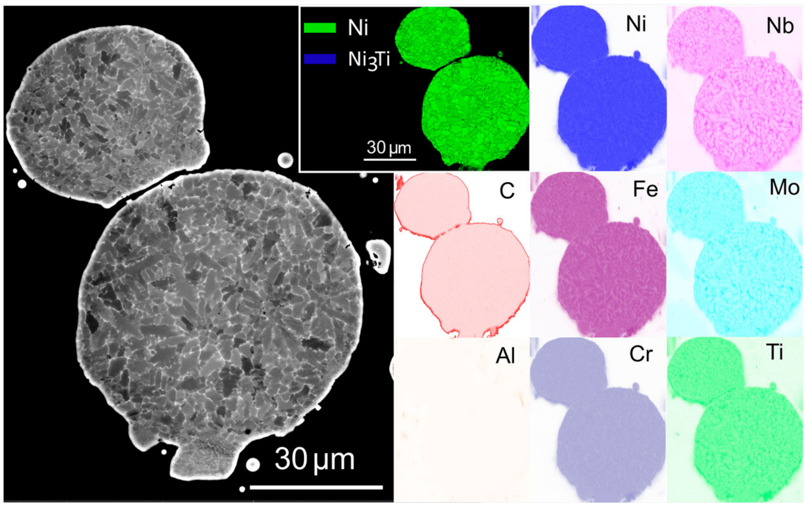

Figure 13. Microstructure of the IN718 gas atomised particles. Backscattered electron (BSE) image (a) shows dendrite structures from the solidification. Electron backscatter diffraction (EBSD) phase map shows nearly single Ni phase in the particles. Energy-dispersive X-ray spectroscopy (EDS) maps reveal segregation of Nb, Mo and Ti as well as depletion of Ni, Fe and Cr at the dendrite boundaries.

has been changed due to a high melt mass flow rate and the assumed injected parent particle size of 4.5 mm is not valid for the calculation anymore. Following this, the gas atomisation process should be further studied with the consideration of different breakup modes for a deeper understanding in the future.

Figure 12(a) and (b) show the gas flow field changing with respect to the changes of the atomising gas pressure and the melt mass flow rates. The case in Figure 12 (b) is assigned with a constant gas mass flow rate, therefore, the GMR decreases with increased melt mass flow rates. It can be seen that with the increases of the atomising gas pressure and GMR, the gas flow velocity in the gas flow field, the size of the high-speed gas jet area, and the relative velocity between the droplets and the gas flow all increase. Combined with the related formulas (Eqs. (15)–(17)) in the secondary breakup model, it can be found that the child droplet diameter is affected by the relative velocity between droplets and gas. With the increase of the relative velocity, the breakup of the droplets is intensified, and the final particle size is reduced. Therefore, the gas atomisation process can be understood better through analysing the characteristics of gas flow field changes.

According to the heat transfer considerations, the cooling rate of the droplets can be estimated based on the empirical equations from [52]. Besides melt superheat temperature and gas temperature, the cooling rate is also strongly dependent on the powder particle size and thermal conductivity. The powder particles within a size range of 20–50 μm are cooled down to 500°C in 5–28 ms. During this cooling time, no γ″(Ni3Nb), γ′(Ni3(Al, Ti,Nb)), or δ (Ni3Nb)-phase formation is expected in the microstructure of the powder particles. The microstructure analysis confirms that the used parameters are in the right range.

Figure 13 shows the microstructure of two powder particles which have diameters close to the D_{50}. The BSE micrograph reveals clear dendritic microstructure from the solidification. Based on the z-contrast shown by the BSE signal, it can be recognised that heavy elements segregate at the dendrite boundaries. EDS maps confirm the segregation of Nb, Mo and Ti and the depletion of Ni, Fe and Cr at the boundaries, which are as expected from the solidification process of the corresponding chemical composition. Please note that the segregation of C at the outer shell of the particles stem from the X-ray signal of conductive embedding materials. Meanwhile, EBSD measurement with a step size of 50 nm, which is approaching the resolution limit of the technique, detected only very limited amount of Ni3Ti-phase appearing at the junctions of the dendrite boundaries beside the matrix Ni-phase.

4. Conclusions

In this study, the droplet breakup, cooling and solidification within the coupling process, the final particle size and distribution, and the influencing factors including the atomising gas pressure and the gas to melt ratio (GMR) on the final particle size were numerically investigated by CFD methods. The simulation results were compared with the experiments conducted with the same process parameters in this work and the data from the literature. The main conclusions are listed in the following.

- For droplet cooling and solidification, the temperature change of the droplets was numerically studied. Through the course of the solidification, the cooling rate of droplets reduces significantly due to the continuous generation of latent heat. As the particle temperature decreases, the heat exchange rate between the particles and the gas phase decreases, resulting in a significant reduction in the cooling rate. The cooling rates of particles with different diameters over time are distinctly different, because small particles have faster heat transfer in the gas flow field than large particles.
- Regarding the droplet breakup behaviour, the droplets break up several times before solidification, therefore the droplet size gradually reduces to form the final particle. For all particles, their solidification time is greater than the breakup time, so that the particles have sufficient time to complete the full breakup behaviour without being affected by the solidification.
- A reasonable D_{50} of the particles was obtained by the numerical simulation, while the simulated particle size distribution is narrower compared to the experimental results, which can be resulted from the pre-set conditions of the injected parent droplets. When both the diameter and the velocity of the injected parent droplets were considered as variables, the simulation result exhibits a wider distribution, which is still comparatively narrower but closer to the experimentally measured values.
- The effects of the GMR on final mean particle size observed in the simulation were studied. An expected change of D_{50} from 53 to 37.5 μm by increasing atomising gas pressure has been well predicted by the simulation. As the melt mass flow rate increases, the GMR decreases and D_{50} increases. However, the D_{50} loses its inverse proportionality to the square root of GMR, when the melt mass flow rate is higher than 0.2 kg/s or lower than 0.1 kg/s. The gas flow velocity in the gas flow field and the size of the high-speed gas jet area increase, as the GMR decreases. Increased gas pressure intensifies the breakup of the droplets and reduces the final mean particle size, as the

relative flow velocity between the gas and the melt is enhanced.

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Notes on contributors

Gezhou Wang graduated from RWTH Aachen University in 2022 with a Master's degree in Automotive Engineering and Transportation. He studied the gas atomization process with numerical methods at the Institute of Materials Applications in Mechanical Engineering in 2022 and then worked on this topic as a student assistant for a few months before entering the industry.

Yuanbin Deng currently leads the research in the process simulation field of the powder metallurgy and ceramics at the Institute for Materials Applications in Mechanical Engineering at RWTH Aachen University. His research interests focus on the development of the digital twin of the powder metallurgical processes from powder to component, for instance sinter-based additive manufacturing and near-net-shape manufacturing by hot isostatic pressing.

Frank Adjei-Kyeremeh is a research assistant in the additive manufacturing group at the Foundry Institute of the RWTH Aachen University.

Jiali Zhang is a postdoctoral researcher at the Institute for Materials Applications in Mechanical Engineering at RWTH Aachen University.

Iris Raffeis is the head of the additive manufacturing group at the Foundry Institute of RWTH Aachen University.

Andreas Bührig-Polaczek is the professor of the Foundry Institute at RWTH Aachen University.

Anke Kaletsch is the head of the division powder metallurgy and ceramics at the Institute for Materials Applications in Mechanical Engineering at RWTH Aachen University. She is also the deputy head of the Institute of Applied Powder Metallurgy and Ceramics (IAPK), which is an associated institute of RWTH Aachen University.

Christoph Broeckmann is the professor of the Institute for Materials Applications in Mechanical Engineering at RWTH Aachen University.

ORCID

Yuanbin Deng http://orcid.org/0000-0002-5057-4876
Frank Adjei-Kyeremeh http://orcid.org/0000-0002-5554-465X
Jiali Zhang http://orcid.org/0000-0002-3544-5910
Iris Raffeis http://orcid.org/0000-0002-2846-5083
Anke Kaletsch http://orcid.org/0000-0001-9356-6932
Christoph Broeckmann http://orcid.org/0000-0002-0790-8148

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